Growth and stem cells

Today I can… ☐ I can distinguish making more cells from cells taking on a specialised job. ☐ I can compare animal growth with plant growth. ☐ I can identify a role for an animal stem cell or a plant meristem.

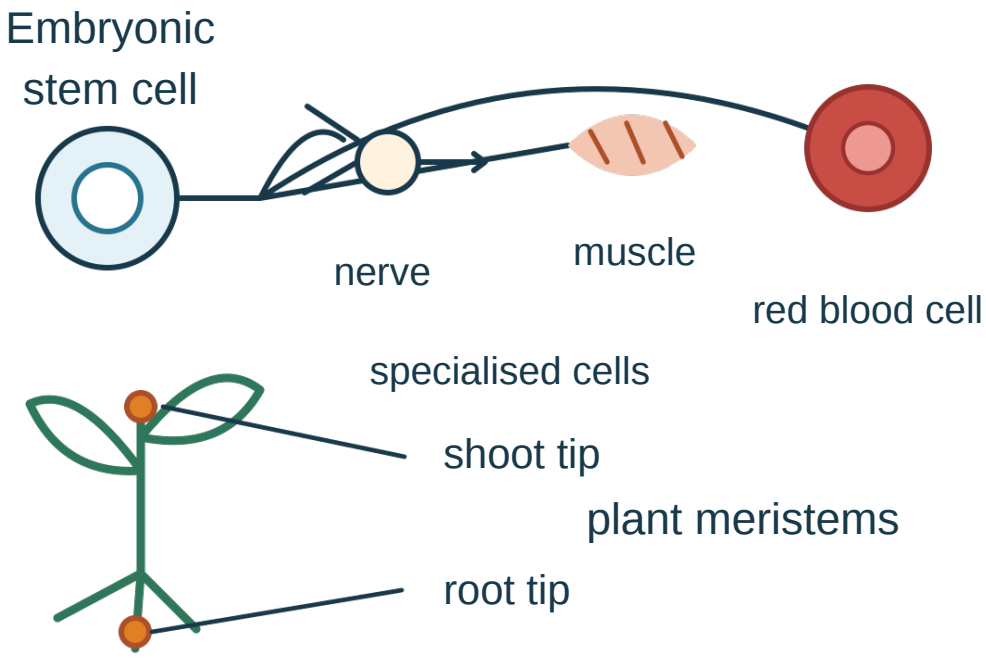
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Knowledge organiser

I am learning to explain how cells divide and then take on different jobs for growth and repair.

Growth and cell jobs

In skin, mitosis adds cells; some stem-cell descendants differentiate for a tissue job. At a shoot tip, meristem cells divide, then elongate and differentiate.



An embryonic stem cell can give rise to different specialised cells; meristems at growing plant tips produce new cells. Simplified; not to scale.

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Knowledge organiser · words and comparison

Word	Meaning
differentiation	a cell becoming specialised for a particular job
stem cell	a cell that can divide and form specialised cell types
meristem	a plant region whose cells divide to make new cells
growth	an increase in size, including by making more cells

Compare the two routes

Skin: mitosis adds cells → differentiation gives a job.

Shoot: meristem adds cells → elongation lengthens cells → differentiation.

Picture colours are illustrative: pale blue stem cell, cream nerve cell, peach muscle cell, red blood cell with a light centre, green plant and orange meristem tips. Colour does not show an observed result.

My language: _____

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